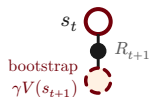
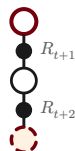
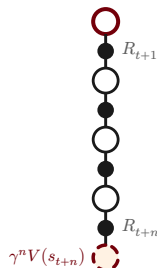
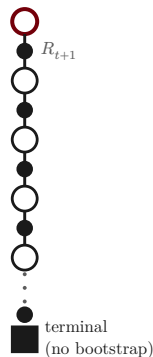


1 sampled reward

TD(0)

2 sampled rewards

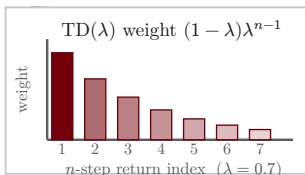
2-step n sampled rewards **n -step**full return G_t **Monte Carlo**depth of backup (shallow $\rightarrow$ deep, bootstrap-free)**TD(0)****Monte Carlo**

variance

high

bias

low



$$n\text{-step target: } G_{t:t+n} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(s_{t+n})$$

$$\lambda\text{-return: } G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n}$$

$$n = 1 \Rightarrow \text{TD}(0); \quad n \rightarrow \infty \Rightarrow \text{Monte Carlo } G_t$$

○ state s ● action a ◌ bootstrap $V(s')$ ■ terminal